

Full length article

Enhancing the confinement of steel reinforced concrete using hexagonal auxetic structures

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ARTICLE INFO

Keywords:

Steel reinforced concrete
Auxetic structure
Beam
Compression
Flexural behavior

ABSTRACT

This study explores an innovative reinforcement approach for steel-reinforced concrete, utilizing auxetic steel structures with various hexagonal shapes to enhance confinement and mechanical performance. Finite element simulations reveal that the auxetic steel structure with the 3D plate model significantly improves ductility and provides robust confinement under compressive loads, outperforming the 3D stick model. In contrast, the 2D auxetic model achieves higher peak stress but exhibits greater brittleness. Furthermore, integrating auxetic and non-auxetic components in reinforced mortar, with auxetic parts in compressive regions and non-auxetic parts in tensile regions, significantly enhances flexural performance by increasing bending resistance and reducing crack propagation. By adjusting the re-entrant angles, these configurations can be optimized for improved mechanical properties. These findings underscore the potential of auxetic materials as valuable complements to traditional reinforcement methods, offering a novel pathway to improve concrete reinforcement strategies.

1. Introduction

The construction industry represents the largest consumer of materials by volume, with concrete being the most widely utilized material. Globally, concrete ranks as the second most consumed substance after water, with an estimated annual consumption of about 30 billion tonnes, and the demand continues to rise [1]. Despite its excellent compressive strength, plain concrete inherently suffers from brittleness and low tensile strength, which limits its application primarily to compression-only scenarios [2,3]. When subjected to load, concrete develops irreversible microcracks, which permit the infiltration of external corrosive agents, thereby compromising its durability and reducing the structure's load-bearing capacity [4,5]. Consequently, concrete relies on reinforcement to attain the ductility and reliability necessary for performance under diverse loading conditions [6–15].

Steel rebar cages provide essential passive confinement through transverse reinforcement, significantly enhancing concrete ductility and delaying failure [16]. This well-established technique improves strength and ductility compared to plain concrete [17–22], though its effectiveness is limited by arching effects that reduce the properly confined concrete volume [16,23,24]. Active confinement techniques applying external pressure before loading show promise for enhanced performance [25], particularly using shape memory alloys [26–28].

However, adoption is hindered by high costs, labor intensity, and prestressing complexity [29].

Auxetic materials exhibit a negative Poisson's ratio, contracting transversely under axial compression and expanding under tension [30–33]. Unlike most natural materials, this behavior enables customized failure modes and enhanced mechanical properties [30]. These properties are achieved through structural reconfiguration, imparting auxetic behavior to diverse materials including textiles, polymers, ceramics, metals, and composites [31,32,34,35].

Recent research shows growing interest in auxetic-cementitious composites for enhanced compressive/flexural properties [34,35]. These systems combine traditional passive confinement with auxetic confinement from embedded architectural patterns [16], significantly improving concrete's mechanical performance. Tzortzinis et al. [16] introduced an approach for reinforcing concrete materials by incorporating 3D re-entrant truss lattice structures. They found that the auxetic lattice provided superior confinement to the mortar matrix, resulting in notable improvements in both strength and ductility when compared to unreinforced mortar specimens subjected to axial compression. Zhong et al. [36] developed a concrete composite featuring an auxetic layered honeycomb structure. Their design demonstrated increased peak stress and maintained stable platform stress when

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<https://doi.org/10.1016/j.tws.2025.113907>

Received 19 May 2025; Received in revised form 14 July 2025; Accepted 27 August 2025

Available online 2 September 2025

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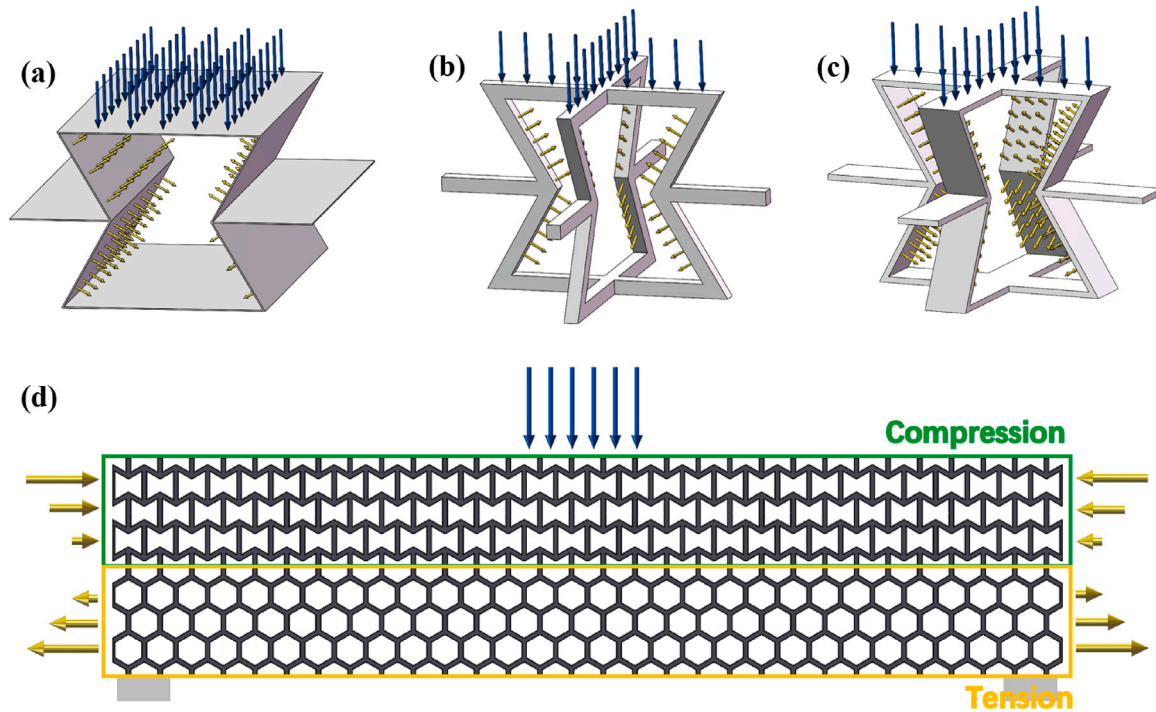


Fig. 1. Schematic representations of the effects of auxetic-enhanced confinement (yellow arrows) on concrete (invisible) subjected to compression loading (blue arrows): (a) 2D, (b) 3D stick, and (c) 3D plate unit cells. (d) Designed hexagonal structure with varied re-entrant angles in the compression and tension regions, intended for embedding in a concrete beam to enhance confinement under bending.

Table 1

Material properties used for FE simulations [39].

Material properties	Steel	Mortar
Young's modulus, E (GPa)	200	4.396
Poisson's ratio, ν (-)	0.3	0.18
Density, ρ (g/cm ³)	7800	2340
Dilation angle, ψ (°)	–	40
Eccentricity, e (-)	–	0.1
σ_{h0}/σ_{c0} (-)	–	1.16
K_c (-)	–	0.666
Viscosity parameter, μ (-)	–	1e-4

subjected to axial compression. In addition, Rosewitz et al. [37] investigated various architected composites made by casting cement mortar with auxetic polymer phases. Their results indicated that the mechanical properties of the architected composite specimens were superior to those of the control samples, attributed to the prevention of localized deformation and failure, which enhanced overall strength. Moreover, Salazar et al. [38] developed 3D-printed polymeric lattices (PLA/ABS materials) to reinforce ultra-high-performance concrete. Through compression and four-point bending tests, they observed that the utilization of these lattices effectively transformed the inherently brittle ultra-high-performance concrete into a ductile material exhibiting strain-hardening behavior, and all flexural specimens exhibited multiple cracking and strain-hardening behaviors until reaching the peak load.

Despite the known benefits of various auxetic structures on mechanical performance, direct comparisons between them when applied to concrete reinforcement are rarely reported. Hexagonal auxetic structures, particularly 2D and 3D stick models (see Fig. 1a and b), are among the most commonly studied for improving concrete reinforcement [34,35]. For a valid comparison, the volumetric relative density of reinforcement must be kept constant. This results in the 2D model having thinner plates than the 3D structure, which inherently makes it weaker. Moreover, the segmentation of concrete by the plates in the

2D model could lead to different failure mechanisms compared to the 3D model. Therefore, a thorough comparison is required to understand how these factors influence the overall performance of auxetic reinforcement across different structural configurations. In addition to 2D and 3D stick models, we also examine a 3D plate model (see Fig. 1c). While the behaviors of the bare 3D plate model have been studied previously [40,41], to the best of our knowledge, its application in concrete reinforcement has not yet been explored.

Additionally, concrete beams are often subjected to bending in practical applications [42,43]. In a concrete beam subjected to flexural loads, the distribution of tension and compression forces varies across the cross-section: the top region undergoes compression, and the bottom region experiences tension, as the bending moment causes the top to shorten and the bottom to elongate (see Fig. 1d). It is important to recognize that auxetic reinforcement affects confinement differently under compression and tension. Specifically, while auxetic structures contribute to enhanced confinement under compression due to their negative Poisson's ratio, they may actually detract from confinement under tensile loads by expanding. Despite this, many studies have applied auxetic structures in both compression and tension regions of concrete beams [37,38]. To optimize the overall flexural performance, it is crucial to consider the force distributions and how auxetic reinforcement interacts with these forces. Improved design strategies for reinforced structures can be developed by accounting for these factors.

The primary objective of this study is to investigate the mechanical performance of hexagonal auxetic structures and their effectiveness in reinforcing concrete. Since concrete is predominantly subjected to both compressive and flexural loads in real-world applications, it is essential to understand how auxetic reinforcement performs under these conditions. Compressive behavior is critical because concrete naturally excels in resisting compressive forces, but auxetic structures may further enhance this by providing superior confinement, leading to improved load-bearing capacity. On the other hand, flexural behavior is equally important because many concrete elements, such as beams and slabs, experience bending stresses in service. By studying the auxetic reinforcement's impact on both compression and tension

Table 2

Ductility indices of the steel reinforced mortar with various hexagonal auxetic structures under compression tests.

Type	D_c	I_5	I_{10}
2D 60°	1.26	3.38	5.16
2D 70°	1.21	3.72	5.93
2D 80°	1.55	4.18	6.28
3D plate 60° 4 mm	2.36	4.55	7.15
3D plate 60° 5 mm	2.33	4.61	7.58
3D plate 60° 6 mm	2.53	4.64	7.65
3D plate 70° 6 mm	2.03	4.58	7.89
3D plate 80° 6 mm	2.02	4.47	7.87
3D stick 60°	1.90	4.57	6.69
3D stick 70°	1.81	4.18	6.45
3D stick 80°	1.81	4.18	6.45

regions during flexure, we can better assess its overall effectiveness. The scope of the research includes examining three hexagonal auxetic structures, namely, 2D, 3D stick, and 3D plate models, using finite element (FE) simulations. We first examine the compressive behaviors of the bare auxetic structures to understand their intrinsic mechanical properties and performance under compressive loads. Subsequently, we evaluate the compressive behaviors of auxetic-reinforced concrete to assess how these structures enhance confinement and load-bearing capacity. Additionally, we explore the flexural behaviors of these three bare auxetic structures, focusing on their response to bending loads and their potential effectiveness in concrete reinforcement. In order to optimize structural performance under flexural stress, we introduce a novel design that combines an auxetic structure in the compression region with a non-auxetic structure in the tensile region (see Fig. 1d). The flexural behaviors of auxetic-reinforced concrete are analyzed to determine the influence of embedded structures on the overall flexural performance. Collectively, the study advances the understanding of auxetic design for concrete reinforcement and offers a pathway toward more efficient structural materials.

2. Methods

2.1. Structure geometries

The re-entrant honeycomb design is achieved by reversing the orientation of the inclined members in a standard honeycomb structure. The 2D version exhibits a negative Poisson's ratio in the vertical direction and one horizontal direction, whereas the 3D version demonstrates a negative Poisson's ratio across all three dimensions. These 2D and 3D hexagonal structures operate via distinct confinement mechanisms (see Fig. 1a and b). Under compression, the 2D model offers a larger contact area with the concrete compared to the 3D stick model, leading to more efficient load transfer. We additionally examine a hybrid 3D plate model (Fig. 1c), combining characteristics of both 2D and 3D stick configurations.

Two types of hexagonal shapes were investigated: an auxetic re-entrant honeycomb (see Fig. 2a) and a non-auxetic honeycomb structure (see Fig. 2b). In the auxetic configuration, the top and bottom horizontal members have lengths of $2l$, while the remaining edges maintain a length of l . The unit cell length, L , is fixed at 20 mm, and the cell width, W , is determined based on the re-entrant angle, α . Unless otherwise specified, α is set at 60° for the auxetic structure. In the non-auxetic configuration, the sizes of central hexagonal edges are the same as m , while the middle horizontal members have lengths of n . Unless otherwise specified, the re-entrant angle, β , is set at 120°. Non-auxetic configuration is considered for the mixed design of the beam (see Fig. 1d). To be compatible with the auxetic configuration, the unit cell sizes, L and W were fixed to be the same as those in the corresponding auxetic configuration, which can be used to determine m and n . The thickness t is determined based on the volumetric relative density of reinforcement ρ^* , which is fixed at 5%. Low relative

densities are commonly used in reinforced concrete civil infrastructure columns, where volumetric reinforcement typically ranges from 3% to 5.5% in non-critical areas, considering both longitudinal and transverse reinforcement [18,39,44].

Three types of models were considered to construct 3-dimensional structures: the 2D model, the 3D stick model, and the 3D plate model, as illustrated in Fig. 1a–c. The 2D model is created by extruding hexagonal shapes and the 3D stick model is constructed by forming hexagonal shapes using sticks with square cross-sectional area, which are then rotated along the symmetrical axis in the vertical direction. These two models are among the most commonly studied for concrete reinforcement [34,35]. In contrast, the 3D plate model is less studied and integrates both extrusion and rotation, combining the features of the 2D and 3D stick models. Note that the plate width (extrusion depth) in 3D plate models can be tuned for better mechanical performance. The size of plate width should be small enough to avoid overlapping during rotation. Three values of plate width are selected accordingly: 4, 5, and 6 mm.

Fig. 2c–e present the reinforcement structures used for the compression tests. Two additional plates, not shown in the figure, are placed at the top and bottom of the reinforcement structure to ensure stability and facilitate the application of compressive loads during testing. The dimensions of the steel and mortar in the x- and z-directions are fixed at 80 mm (corresponding to four repeating units). The length in the compression (y-) direction ranges from 70 mm to 80 mm, depending on the re-entrant angle (*i.e.*, seven repeating units in the y-direction). Fig. 2f depicts the setup and geometric parameters for the four-point bending test. The mortar beam has a volume of 700 mm × 80 mm × 80 mm, with the steel reinforcement structure embedded inside. The number of repeating units of steel structures in the x-direction is 60. Two types of steel reinforcement structures are considered: one composed entirely of auxetic units, and the other combining auxetic units in the compression region with non-auxetic units in the tension region, as shown in Fig. 1d. This hybrid design aims to optimize structural performance under flexural loading conditions as described above.

2.2. Finite element model

To effectively simulate the failure modes and mechanical performance of auxetic structures under compressive and flexural loading, the FE models were developed. In the compression and bending tests, the bottom rigid supports were fully constrained at all degrees of freedom at the reference nodes. The top rigid loading cells were restricted to downward movement along the y-axis at a constant velocity of 0.2 mm/s. The steel structures were constructed as a 3D assembly and embedded into the mortar solid elements using the embedded element technique. The general contact algorithm was employed to automatically account for all potential contact interactions. The penalty contact formulation was applied to impose contact constraints between surfaces, with the penalty stiffness for the “hard” contact option being automatically determined by the software. To account for frictional forces between the contacting surfaces, the Coulomb friction model was utilized, with a friction coefficient set to 0.2.

The 2D and 3D plate structures were simulated using S4R shell elements with reduced integration and a default hourglass control algorithm, while the 3D stick structures were constructed with B31 beam elements, which account for transverse shear deformation. The mortar was modeled using C3D8R solid elements, in alignment with previous studies that have employed this formulation for similar materials and structural configurations [16,36,39]. The choice of reduced integration (R) formulation was made to mitigate common issues such as shear locking and volumetric locking, which can arise in standard fully integrated hexahedral elements when modeling nearly incompressible materials or bending-dominated problems. The C3D8R element includes three translational degrees of freedom per node, and although it

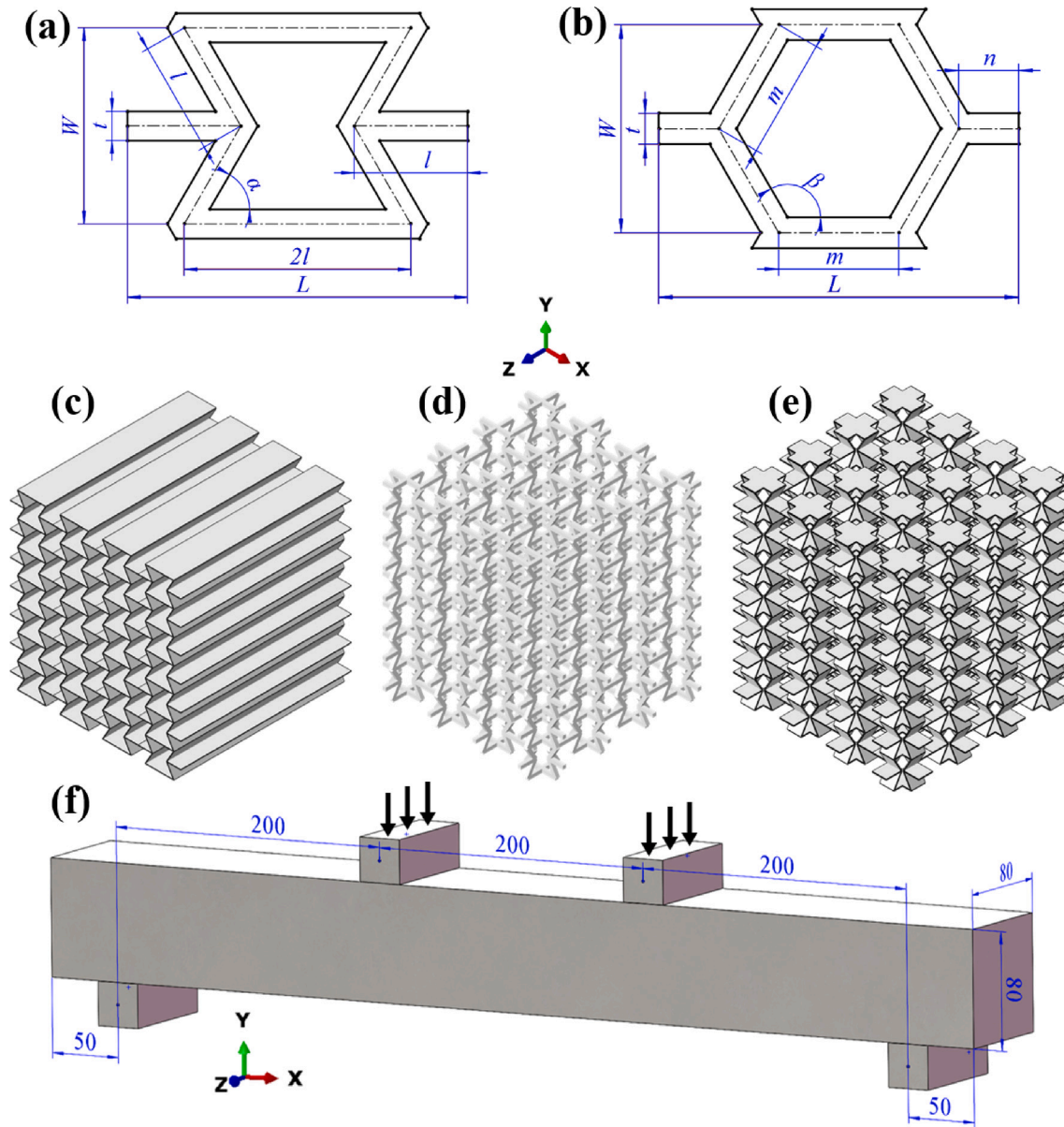


Fig. 2. Geometric parameters for hexagonal (a) auxetic and (b) non-auxetic shapes. Steel structures for compression tests with (c) 2D, (d) 3D stick, and (e) 3D plate models. (f) Geometric configuration for four-point bending tests.

does not explicitly account for rotational degrees of freedom, the mechanical behavior of the mortar matrix under compression and bending tests in our simulations was primarily governed by axial deformation and distributed bending, both of which are sufficiently represented through translational displacements. Convergence analysis with varying mesh sizes was conducted for the compression simulation of the 2D plate reinforcement structure embedded in mortar, as shown in Fig. S1. By comparing the peak stress and plateau stress across three different mesh sizes, convergence was observed by refining meshes. As the deviations between the models with mesh sizes of 1 mm and 1.5 mm were less than 10%, a 1.5 mm mesh size was selected for all numerical simulations to balance computational efficiency and accuracy.

The material properties are taken from Ref. [39] and given in Table 1. To simulate the plastic properties of the mortar, a concrete damage plasticity model was utilized. Additionally, the effective plastic response of the steel was defined using a stress-strain relationship: starting with an initial yield stress of 850 MPa at zero plastic strain, rising to 1050 MPa at a plastic strain of 0.06, then decreasing back to 850 MPa at 0.085 plastic strain, and eventually reducing to 1 MPa at

0.135 plastic strain, representing the loss in load-bearing capacity due to fracture.

2.3. Validations

The FE models developed in this study have been validated against the deformation responses of auxetic structures and composites reported in the literature, both from experimental data and prior simulations [39,45,46]. Validation was conducted for three distinct cases: (1) dynamic crushing of a bare 3D stick model, (2) quasi-static compression of a steel-reinforced mortar incorporating a 3D stick model, and (3) four-point bending of concrete reinforced with steel rebars. Key simulation parameters for the validation tests were adopted from the original references, ensuring consistency where possible. As shown in Fig. S2, the stress-strain and stress-displacement curves from our simulations exhibit good agreement with both experimental and numerical results from the literature. Minor discrepancies between the simulation results are observed, primarily due to certain unreported parameters in the original studies, which were approximated using values from references

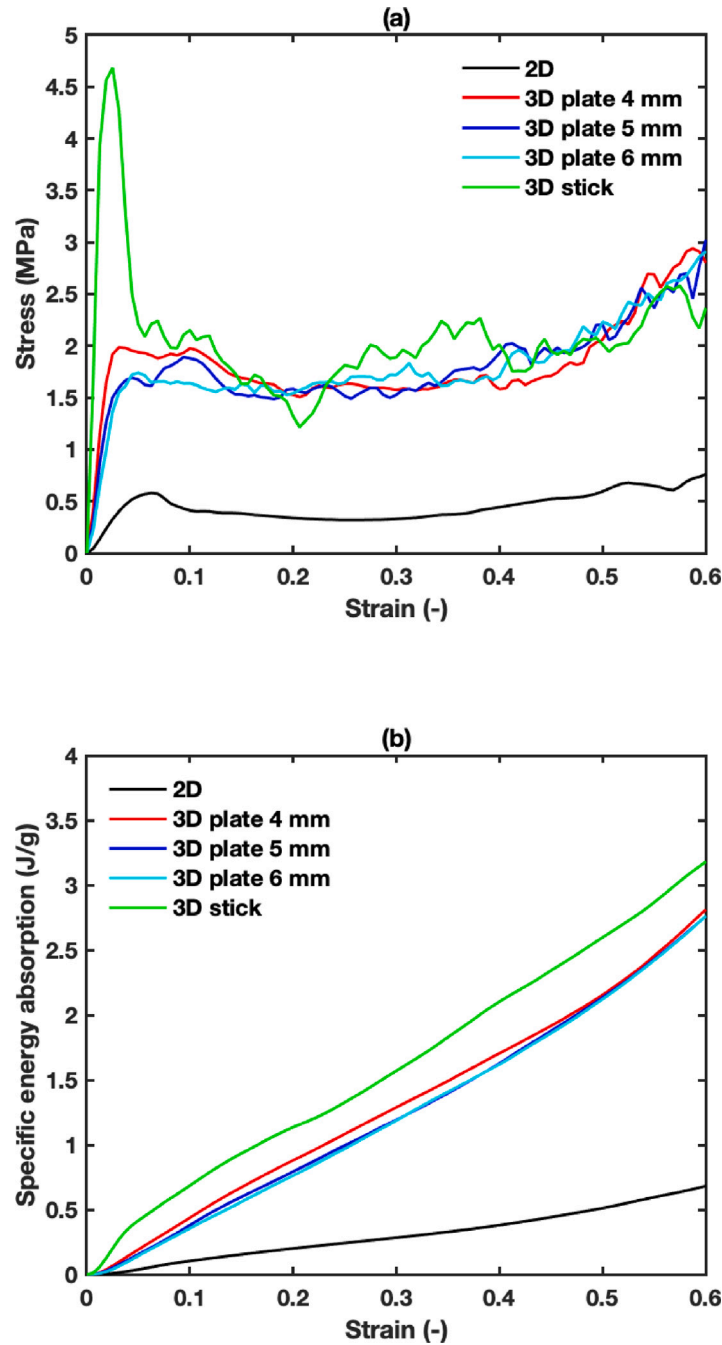


Fig. 3. (a) Stress–strain curves from compression tests of bare steel structures for various models, and (b) the corresponding specific energy absorption as a function of strain.

with comparable conditions. Remarkably, in Ref. [39], a C3D10 solid element was employed to model the steel component. The B31 beam elements were selected for their computational efficiency in modeling slender lattice components, aligning with established methodologies for auxetic structures [47–49]. Although C3D10 solid elements offer higher fidelity in capturing local stress distributions, B31 elements provide adequate accuracy in representing global mechanical responses – such as load–displacement behavior and energy absorption – while significantly reducing computational cost. This advantage is particularly important for the large-scale parametric analyses conducted in this study. A comparison with simulations using C3D10 elements (Fig. S2b) reveals similar trends in the stress–strain response, with a peak stress discrepancy of approximately 16 MPa (~10%). In contrast, the variation observed in repeated experiments is around 26 MPa,

and both simulation approaches yield results within the experimental range. Consequently, the beam-element formulation demonstrates reliable predictive capability without introducing qualitative deviations from experimental observations.

3. Results and discussion

3.1. Compressive behaviors of bare auxetic structures

Fig. 3a presents the stress–strain curves obtained from compression tests of various bare steel structures, ensuring constant steel volume across all models. The overall behavior of these curves aligns well with results reported in the literature [45,50,51]. The compression process initiates with a linear elastic deformation phase, followed by a stress

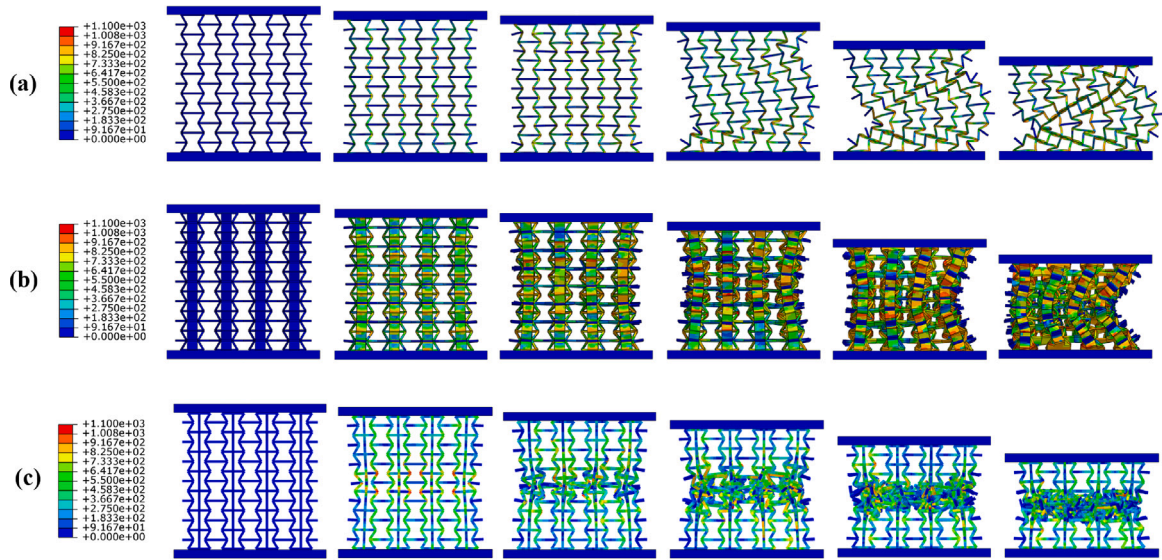


Fig. 4. Von-Mises stress maps for compression tests of bare structures with a re-entrant angle of 60°: (a) 2D hexagonal model, (b) 3D plate model with a plate width of 4 mm, (c) 3D plate model with a plate width of 6 mm, and (d) 3D stick model. Each row shows strain increments of 0, 0.03, 0.06, 0.125, 0.25, and 0.375, respectively, from left to right.

drop after reaching the first peak. This leads to a plateau stage, where the structures exhibit auxetic deformation characteristics. At very high strain levels, the re-entrant structures undergo densification, resulting in a sharp increase in stress. Nevertheless, significant differences are observed between the various models. The peak stress of the 3D stick model is considerably higher than that of the 3D plate models, while the 2D model exhibits the lowest peak stress. The plateau stress values of the 3D stick and 3D plate models are comparable at around 1.7 MPa and significantly exceed that of the 2D model at around 0.4 MPa. Increasing the plate width in the 3D plate model results in a slight reduction in the stress curves. These variations are primarily attributed to the thickness parameter, t (see Fig. 2a and b), as thinner structures are more susceptible to deformation under stress. Since the steel volume is constant, the 3D stick model exhibits the largest t , while the 2D model has the smallest. Additionally, increasing the plate width in the 3D plate model reduces the value of t .

To analyze deformation modes, Von-Mises stress distributions are illustrated in Fig. 4. In the 2D model (Fig. 4a), stress localizes primarily around the joint points, notably those positioned two to three layers from the substrate. Initial deformation emerges at the top and bottom regions, and at higher strain levels (e.g., exceeding 0.125), the plates exhibit a rightward buckling response. In contrast, the 3D stick model displays a distinctly different deformation pattern (Fig. 4c), where stress concentrates at the central joint points as compression initiates deformation. The vertical struts buckle inward, with struts on the right side bending leftward and those on the left bending rightward, a buckling behavior previously noted by Teng et al. [50]. The 3D plate model integrates characteristics from both the 2D and 3D stick models (Fig. 4b). Stress primarily concentrates around the joint points, though it exhibits a more uniform distribution overall. Under low strain conditions, the vertical plates buckle inward; at higher strain levels, however, this buckling shifts to a leftward direction.

To evaluate the efficiency of energy absorption, the specific energy absorption (SEA) is examined, which quantifies the energy absorbed per unit of mass. SEA is defined as the ratio of the total energy absorbed by the structure to its total mass:

$$SEA = \frac{E_a}{m}, \quad (1)$$

where E_a represents the total absorbed energy, which can be expressed as:

$$E_a = \int_0^{\delta} F d\delta, \quad (2)$$

with δ being the displacement and F the corresponding force applied to the structure. Fig. 3b displays the SEA values as functions of strain. The SEA values of the 3D stick model are significantly greater than those of the 3D plate models, whereas the 2D model shows the lowest SEA .

Figs. S3a-c illustrate the influence of the re-entrant angle α on the stress-strain curves from compression tests of bare steel structures. Overall, increasing α from 60° to 90° results in a notable rise in stress, particularly at the peak stress values. This behavior can be attributed to the fact that as α deviates from 90°, a greater portion of the force is distributed laterally, resulting in more efficient load transfer and enhanced deformation along the compression axis. The corresponding SEA results are given in Figs. S3d-f. Consistent with the preceding findings, the 3D stick model exhibits significantly higher SEA values compared to the 3D plate models, while the 2D model consistently demonstrates the lowest SEA performance. Furthermore, the results indicate that increasing the parameter α enhances the energy absorption capacity of the bare structures, highlighting a positive correlation between α and structural energy dissipation efficiency. Fig. S4 displays the Von-Mises stress maps for compression of bare structures with a re-entrant angle of 80°. Overall, the stress patterns are consistent with those shown in Fig. 4. Notably, the 3D plate model with a re-entrant angle of 80° exhibits an inward buckling mode, closely resembling the deformation behavior of the 3D stick model. In contrast, when the re-entrant angle is reduced to 60°, the 3D plate model displays a leftward buckling mode, which is more comparable to the response observed in the 2D model. These results indicate that the buckling mode of bare 3D plate structures is strongly influenced by the re-entrant angle.

3.2. Compressive behaviors of auxetic reinforced mortar

Fig. 5a presents the stress-strain responses from uniaxial compression tests of mortar specimens reinforced with steel of various structural configurations. Note that the peak stress of pure mortar is at 43.29 MPa with a strain value of 0.015. The composite results significantly improved compared with pure mortar or bare steel structures (cf. Figs. 5a and 3a). The 2D model shows a brittle failure mode with a strain value of 0.038. In contrast, all 3D-reinforced mortar specimens exhibit a similar pattern, with a sharp increase in the peak stress, followed by a gradual decline. The 2D model achieves the highest peak stress of around 140 MPa, whereas the 3D stick model shows the lowest

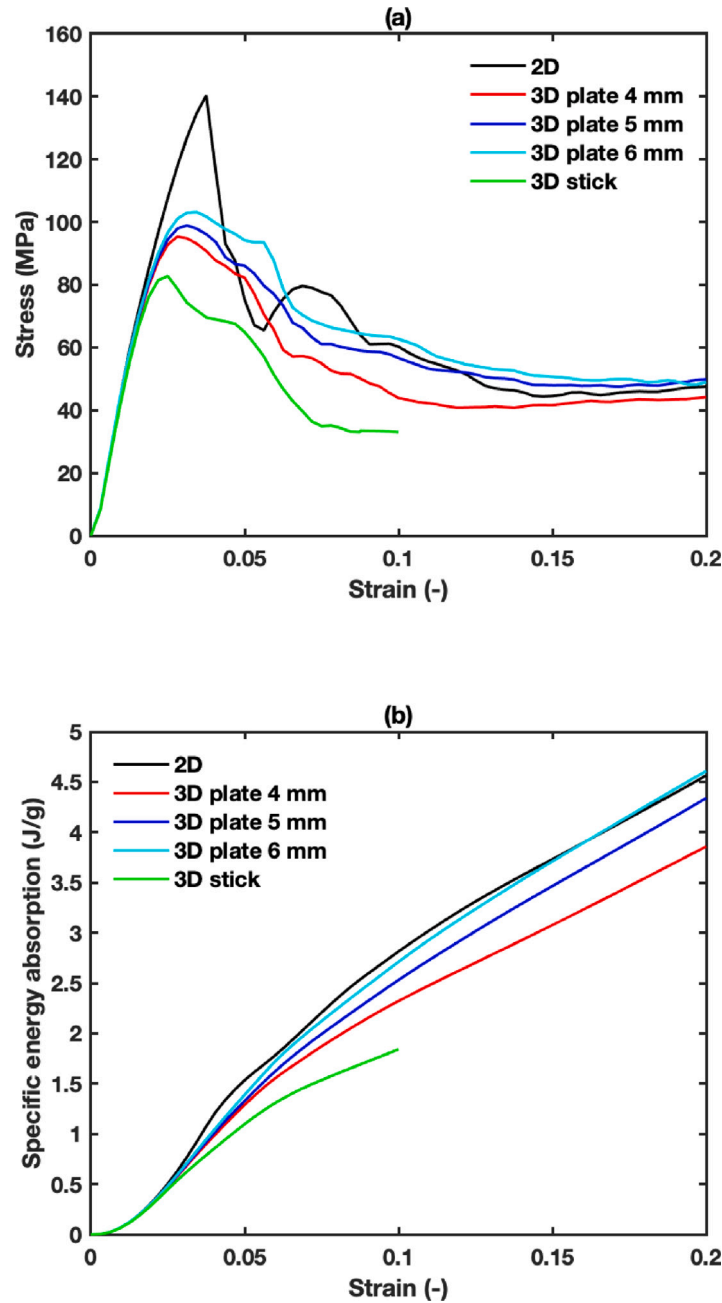


Fig. 5. (a) Stress–strain curves from compression tests of mortar reinforced with steel for various models, and (b) the corresponding specific energy absorption as a function of strain.

peak stress at approximately 82.8 MPa at a strain of 0.025. Among the 3D plate models, peak stress and corresponding strain values increase with plate width. Notably, plate width is limited to prevent overlapping during rotation (refer to Fig. 1c). At the maximum plate width of 6 mm, the peak stress reaches 103.2 MPa with a strain of 0.034.

Fig. 5b illustrates the specific energy absorption results for each reinforcement configuration. The 2D model and the 3D plate model with a plate width of 6 mm show comparable energy absorption capacities, outperforming other configurations. While the 2D model achieves a notably higher peak stress, its brittle failure mode leads to an energy absorption performance similar to that of the 3D plate model.

Fig. 6 presents the Von-Mises stress distributions in steel structures embedded within mortar under uniaxial compression. The deformation modes exhibit more lateral expansion and differ markedly from cases without mortar, as the void spaces are now filled with mortar, which

alters the load distribution and failure mechanisms (*cf.* Figs. 4 and 6). The blue zones with significant deformations reveal elements that have surpassed peak strength within the plastic range, entering a softening phase preceding fracture. In the 2D model (Fig. 6a), the failure pattern exhibits triangular end cracking, aligning with previous findings on 2D auxetic structure-reinforced composites made from cement mortar cast around an acrylonitrile butadiene styrene polymer phase.[37] Conversely, in 3D models (Fig. 6b–d), initial failure tends to initiate randomly near the central region of the composite. The first fractures predominantly occur along horizontal elements, with damage progressing along adjacent horizontal struts surrounding the initially fractured area.

To quantify the ductility of the composite, the approach outlined by Foster and Attard [52] was applied to determine the ductility indices D_c , I_5 , and I_{10} . The yield strain, serving as a reference for these indices,

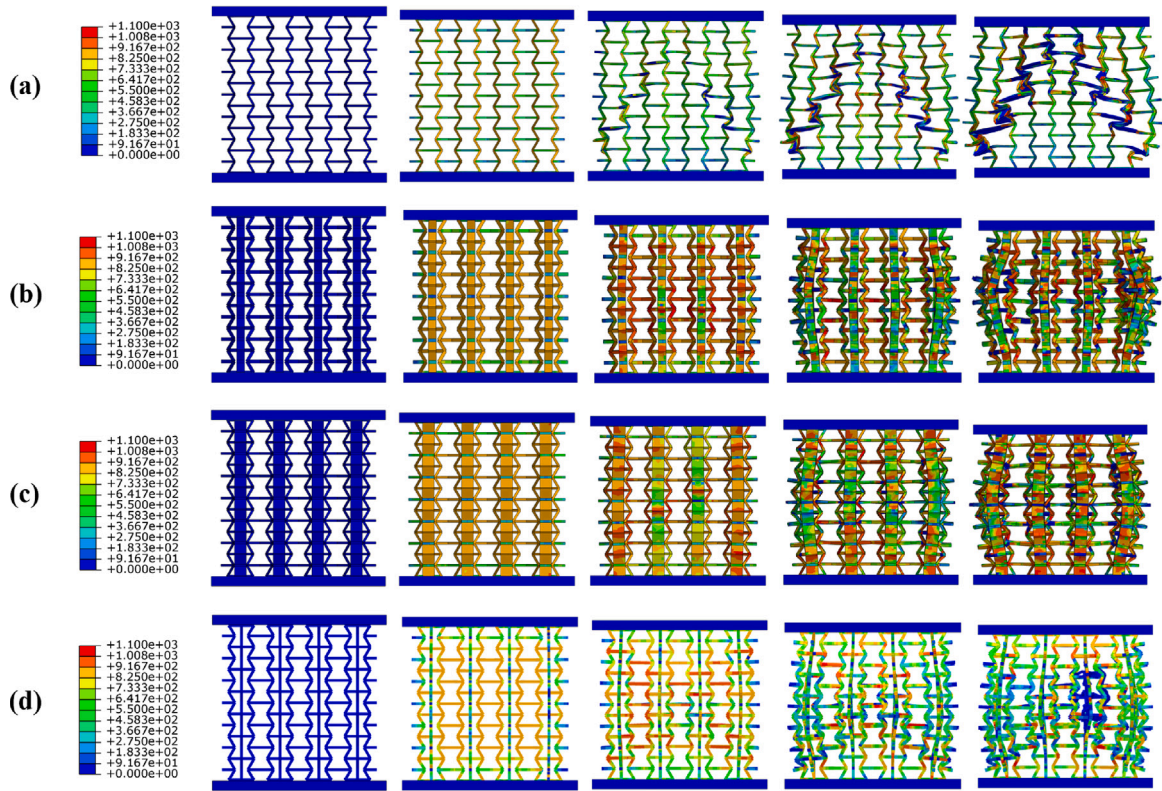


Fig. 6. Von-Mises stress maps for compression tests of steel structures with a re-entrant angle of 60° embedded in mortar: (a) 2D hexagonal model, (b) 3D plate model with a plate width of 4 mm, (c) 3D plate model with a plate width of 6 mm, and (d) 3D stick model. Each row shows strain increments of 0, 0.025, 0.05, 0.075, and 0.1, respectively, from left to right. The plate thickness of 2D and 3D plate models are magnified by 5 and 3, respectively.

was determined using the 3/4 method, in which a line is extended from the graph's origin through the point corresponding to $0.75P_{\max}$ up to $P_{\max}$. The combined ductility ratio D_c is then calculated by taking the strain at which 85% of the maximum strength is retained in the post-peak phase and dividing it by the yield strain (Eq. (3)). This ratio serves as a measure of the structural member's ductility near peak performance. Additionally, the energy ductility indices I_5 and I_{10} are computed by dividing the area A under the force-strain curve at $3\times\epsilon_y$ by the area under the curve up to ϵ_y for I_5 , and by dividing the area under the curve up to $5.5\times\epsilon_y$ by the area up to ϵ_y for I_{10} (Eqs. (4) and (5)). The ductility index definition utilized in calculating the energy ductility metrics relies on the area under the curve of axial load N against axial strain ϵ_{axial} , enabling the inclusion of residual capacity in the post-peak phase.

$$D_c = \frac{\epsilon_{0.85}}{\epsilon_y}. \quad (3)$$

$$I_5 = \frac{A_{3y}}{A_y}. \quad (4)$$

$$I_{10} = \frac{A_{5.5y}}{A_y}. \quad (5)$$

Table 2 presents the ductility indices of the composite with various models. It is evident that the 3D plate model exhibits the highest ductility index, followed by the 3D stick model with a moderate value, while the 2D model shows the lowest ductility index. Increasing the plate width in the 3D plate model further enhances the ductility indices. This ductile behavior can be attributed to the distribution of stress within the composite material. In the 3D plate model, the contact area between the steel and mortar is significantly greater than in the 3D stick model, promoting a more uniform stress distribution. In contrast, the 2D model features a separation of mortar by steel plates, limiting stress distribution efficiency.

Fig. 7 shows the effects of re-entrant angle α on the mechanical performance of composites under uniaxial compression. As expected, when α reaches 90° , the mechanical performance of all models decreases substantially as the structure transitions to a non-auxetic configuration. This observation underscores the critical role of the auxetic-induced confinement effect in enhancing mechanical performance. In the 2D model (see Fig. 7a), the stress-strain response shows a moderate decrease as α increases from 60° to 70° , due to the reduced transfer of vertical compression load to the lateral direction, which in turn weakens the auxetic-induced confinement effect. However, at $\alpha = 80^\circ$, a notable increase in peak stress and ductility is observed, indicating enhanced structural performance. In contrast, in the 3D plate model (see Fig. 7b), both peak stress and the corresponding strain decrease consistently as α increases. Interestingly, the 3D stick model shows minimal sensitivity to α across the range of 60° to 80° , with only minor variations in mechanical response (see Fig. 7c). This limited influence of α on the 3D stick model is likely due to reduced interaction between mortar and steel, resulting from the smaller contact area. Additionally, the specific energy absorption results are given in Fig. 7d–f. The energy absorption capacity is similar between the 2D and 3D plate models, while the 3D stick model exhibits the lowest capacity.

Fig. S5 presents the Von-Mises stress distributions in steel structures (with $\alpha = 80^\circ$) embedded within mortar under uniaxial compression. It can be seen that more vertical elements fail at larger α by comparing Fig. S5 and Fig. 6. Moreover, the lateral swelling of the steel structure starts at smaller strain conditions with larger α . Remarkably, at $\alpha = 80^\circ$, the failure pattern in the 2D model exhibits rectangular end cracking, contrasting with the triangular end cracking observed at $\alpha = 60^\circ$. This difference likely contributes to the enhanced mechanical behavior of the 2D model at $\alpha = 80^\circ$, as rectangular end cracking tends to preserve a larger portion of the undamaged composite for bearing the load and delaying failure. Furthermore, the effects of α on ductility indices are summarized in Table 2. Generally, as α increases from 60° to 80° , the

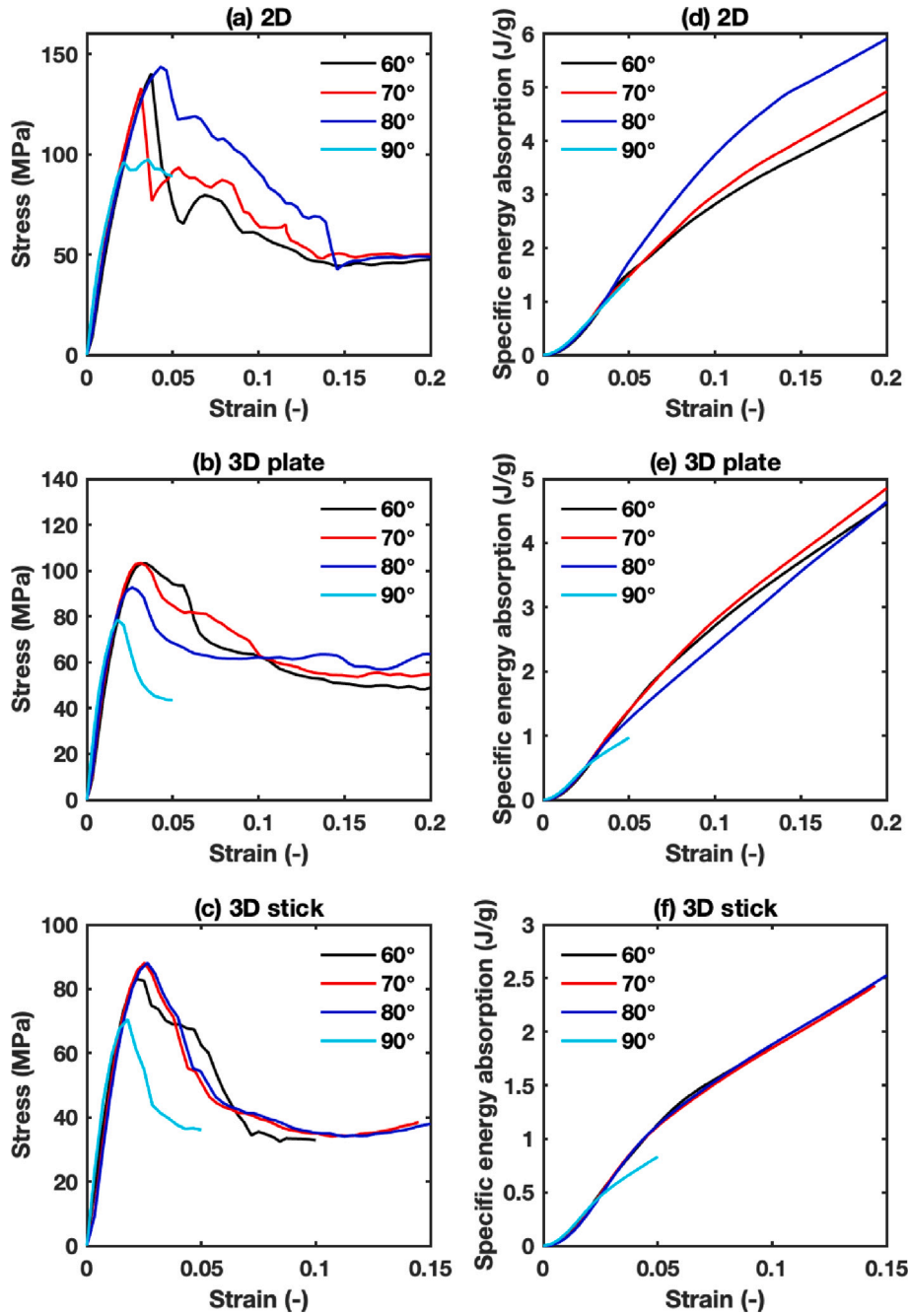


Fig. 7. Stress–strain curves from compression tests of mortar reinforced with steel structures featuring varying re-entrant angles for: (a) 2D models, (b) 3D plate models (6 mm plate width), and (c) 3D stick models. The corresponding specific energy absorption as a function of strain for: (d) 2D models, (e) 3D plate models, and (f) 3D stick models.

composites exhibit increased brittleness. However, in the 2D model, the case with $\alpha = 80^\circ$ demonstrates the best ductile behavior, which can also be attributed to the alteration in deformation modes.

3.3. Flexural behaviors of bare auxetic structures

Fig. 8a–c present the stress–displacement curves from four-point bending tests conducted on bare steel structures. These include structures incorporating exclusively auxetic elements and mixed designs where steel regions integrate both auxetic and non-auxetic configurations with various re-entrant angle β (see Fig. 1d). The corresponding results for specific energy adsorption are given in Fig. 8d–f. For the 2D and 3D plate models, the load increases monotonically with displacement. In contrast, the 3D stick model exhibits an initial increase in load,

followed by fluctuations. Under identical displacement conditions, the 3D stick model exhibits the highest load-bearing and energy absorption capacity, followed by the 3D plate model, while the 2D model sustains the lowest load. The mixed design demonstrates superior load-bearing capacity compared to the purely auxetic design. Additionally, as the angle parameter, β , increases from 110° to 130° , the load-bearing capacity correspondingly improves. The corresponding Von Mises stress maps are presented in Figs. S6–S11. These results reveal distinct deformation characteristics among the different structural models. In both the bare 2D and 3D plate models, the stress maps indicate a more global deformation pattern, with stress distributed broadly across the structure. This suggests a more uniform load transfer and a higher degree of structural continuity. In contrast, the 3D stick model exhibits

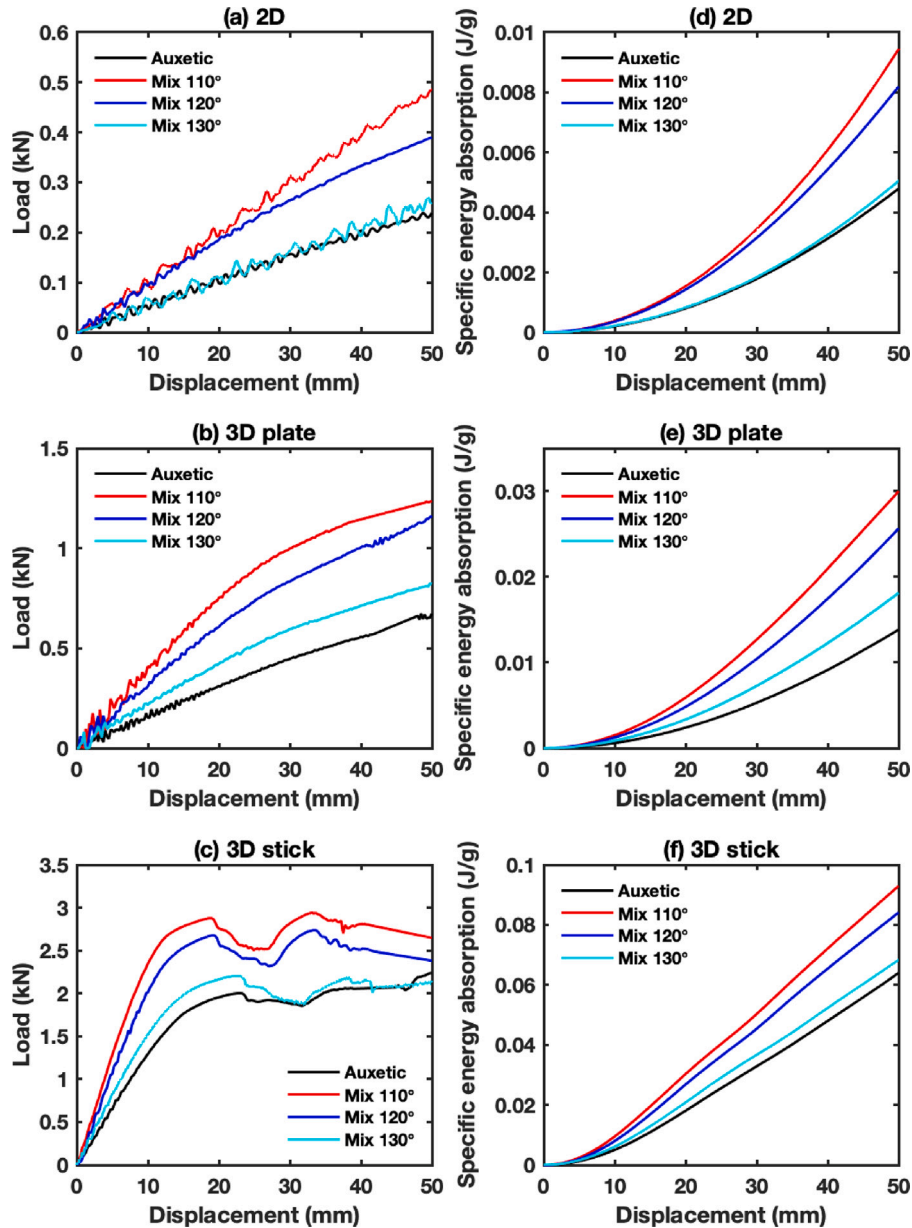


Fig. 8. Stress-displacement curves from four-point bending tests of bare steel structures with varying re-entrant angles for: (a) 2D models, (b) 3D plate models, and (c) 3D stick models. The corresponding specific energy absorption as a function of strain for: (d) 2D models, (e) 3D plate models, and (f) 3D stick models.

highly localized deformation, with stress concentrations primarily confined to the regions adjacent to the two loading cells. This localized response indicates that the stick-based framework is more prone to stress accumulation near points of load application, potentially leading to early failure in these zones. These differences highlight the critical role of structural geometry in governing deformation modes and stress distribution under flexural loading.

3.4. Flexural behaviors of auxetic reinforced mortar

Fig. 9a–c display the stress-displacement curves obtained from four-point bending tests on mortar specimens reinforced with steel structures. The reinforcement designs include those composed solely of auxetic elements, as well as mixed designs that combine auxetic and non-auxetic regions, each with different re-entrant angles, β . For purely auxetic designs, the 3D plate model demonstrates the highest load-bearing capacity, followed by the 3D stick model, with the 2D model exhibiting the lowest capacity. For example, at a displacement

of 32 mm, the loads sustained by the 2D, 3D plate, and 3D stick models are approximately 15, 37, and 20 kN, respectively.

Importantly, compared with the pure auxetic design, our findings reveal that the mixed design notably enhances the load-bearing capacity of steel-reinforced mortar beams, with this improvement strongly influenced by the angle parameter, β . As β increases from 110° to 130°, the beam's load-bearing capacity correspondingly improves. Among the three models, the 2D model exhibits the greatest sensitivity to changes in β . For instance, at a displacement of 32 mm, a beam with a mixed design and $\beta = 110^\circ$ can sustain approximately 47 kN, compared to around 27 kN when β is increased to 130°. In contrast, the load-displacement curves for the 3D plate and 3D stick models display a weaker dependence on β , resulting in similar load-bearing capacities between the 2D and 3D plate cases when $\beta = 110^\circ$. Additionally, it is observed that the mixed design increases the brittleness of the beam, with ductility decreasing as β rises. Moreover, Fig. 9d–f present the specific energy absorption curves. While the mixed design improves load-bearing capacity, the maximum specific energy absorption

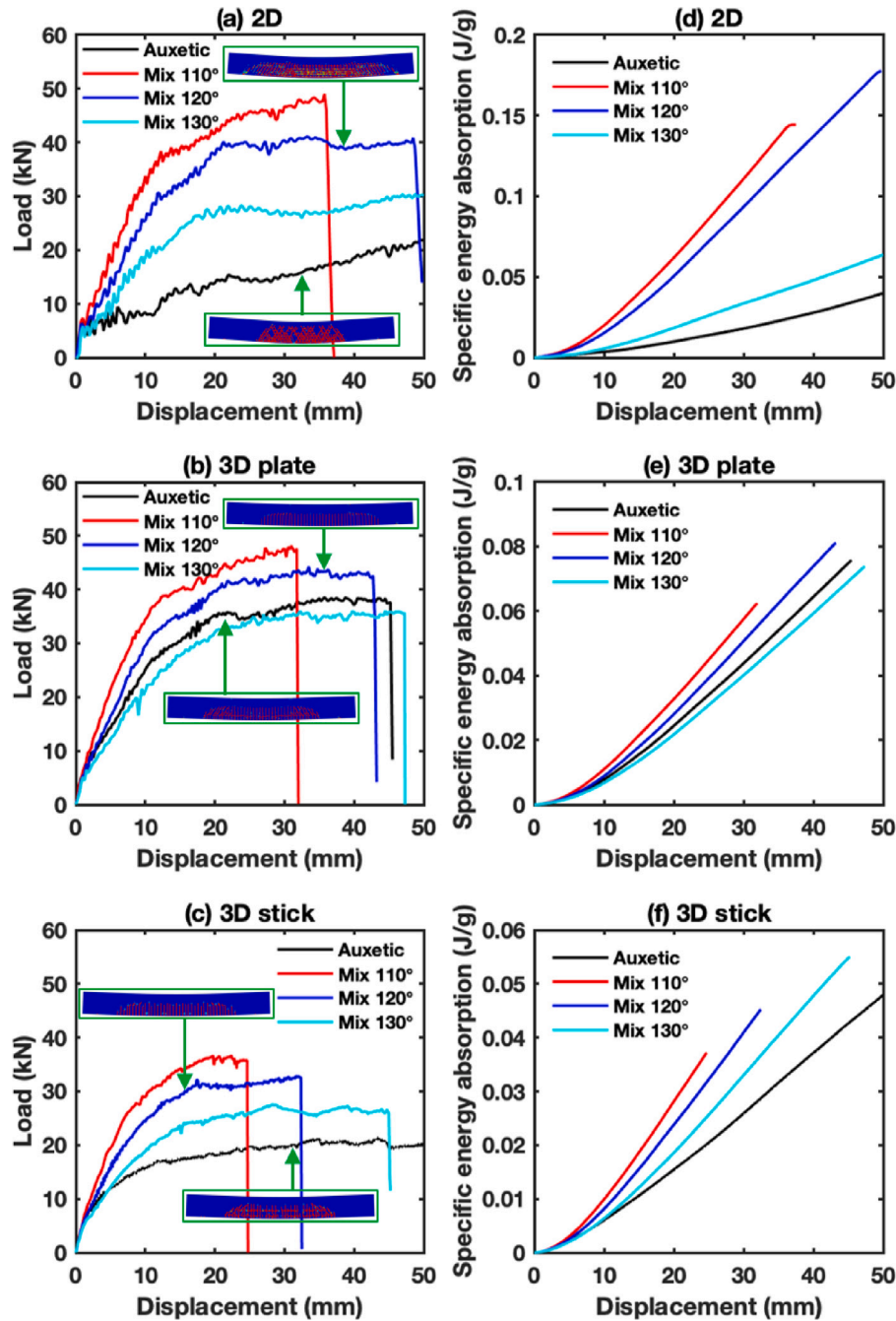


Fig. 9. Stress–displacement curves from four-point bending tests of mortar reinforced with steel structures featuring different auxetic design and re-entrant angles for: (a) 2D models, (b) 3D plate models, and (c) 3D stick models. The corresponding specific energy absorption as a function of strain for: (d) 2D models, (e) 3D plate models, and (f) 3D stick models. The inset contour figures illustrate the tension damage parameter for the reinforced mortar under bending at a displacement of 12 mm.

decreases due to reduced ductility. Note that Rosewitz et al. [37] previously investigated the flexural performance of mortar reinforced with 2D auxetic polymer structures. Their experimental results demonstrated that incorporating an auxetic polymer phase significantly enhanced the flexural strength and toughness of the mortar compared to unreinforced plain white mortar. Our findings reveal that these flexural properties can be further optimized by incorporating a hybrid design on the reinforcing structure, providing an additional improvement in mechanical performance.

Figs. S12–S14 give Von-Mises stress maps of the steel structures with pure auxetic design or mixed design embedded in mortar. It is observed that stress concentrations are primarily localized in the top

and bottom regions near the compressive cells, while minimal stress is observed in the second row of repeating units. If additional vertical units are incorporated beyond the current configuration of four, a non-symmetrical distribution of auxetic and non-auxetic structures could be applied, potentially optimizing load distribution and improving mechanical performance.

Figs. S15–S17 and Supplementary Movies 1–6 show evolution of the tension damage parameter of the mortar with various steel models embedded inside. In addition, inset contour figures in Fig. 9 are extracted from these figures to show detailed views at a 12 mm displacement. The tensile damage patterns in mortar exhibit notable variations across different models. In the 2D auxetic design, failure initiates at the central

region and propagates toward both ends. The cracks form a cross-shaped pattern, aligning with the orientation of the inclined plates in the steel structure. However, in the auxetic designs of the 3D plate and 3D stick models, damage patterns initiate as distinct vertical paths distributed separately within the central region. As displacement increases, horizontal cracks emerge, progressively linking the vertical cracks. Notably, in the 3D plate model, cracks are finer and more densely distributed compared to those in the 3D stick model.

Notably, mixed designs notably alter damage patterns, resulting in delayed failure and generally thinner cracks. In the 2D mixed design, cracks propagate in a more parallel pattern compared to the cross-shaped cracks observed in the 2D auxetic design. In the 3D plate and stick models with mixed designs, horizontal crack formation is significantly restricted. This restriction likely contributes to the observed increase in load-bearing capacity, as cross-shaped cracks are more susceptible to further propagation and potential composite failure.

This study examined a novel approach for enhancing the confinement and mechanical performance of steel-reinforced concrete by incorporating hexagonal auxetic structures. By employing both 2D and 3D auxetic configurations, specifically, stick and plate models, this work advances the field of concrete reinforcement, where traditional materials have long prevailed. A key finding is the distinct confinement mechanisms provided by 2D and 3D auxetic structures: the 3D plate structures exhibit superior confinement under compression, significantly enhancing mortar ductility due to their capacity for more effective confinement of the mortar. In contrast, the 2D auxetic model demonstrates higher peak stress but exhibits increased brittleness. This differentiation underscores the potential to tailor auxetic reinforcement configurations to specific structural demands. Furthermore, the integration of auxetic structures with non-auxetic parts for reinforcing mortar has been shown to enhance the flexural performance of reinforced mortar. By combining auxetic and non-auxetic elements, this study reveals a promising approach for customizing reinforcement properties to achieve high load-bearing capacity. This combination is particularly advantageous in flexural applications, where the mixed design enables higher bending resistance while allowing for controlled deformation, ultimately leading to a reduction in crack propagation and an increase in overall lifespan. Such insights provide a foundation for future design strategies that aim to optimize the placement and configuration of auxetic and traditional reinforcements in concrete. The impact of this study on the field of concrete reinforcement and structural engineering is multifaceted. First, this research highlights the application of auxetic materials as a viable alternative or complement to conventional steel reinforcement, addressing limitations in compressive and flexural strength that have constrained reinforced concrete structures. This study demonstrates the confinement and mechanical benefits of auxetic structures, providing a clear foundation for using auxetic designs in civil and architectural applications. Secondly, this work advances the development of customized reinforcement strategies, particularly those focused on resilience to complex stress conditions. The enhanced confinement provided by auxetic structures opens new avenues for achieving more robust infrastructures, supporting the construction of structures that can withstand increasingly demanding operational environments.

4. Conclusion

This study presents a comprehensive investigation into the mechanical behaviors of mortar reinforced with various auxetic structures under compression and flexural loading utilizing finite element simulations. The results demonstrate the significant potential of auxetic materials to enhance the performance of conventional reinforced concrete, addressing limitations typically associated with traditional reinforcement strategies. The key findings and contributions are summarized as follows:

- The 3D auxetic plate models exhibit superior confinement and ductility under compression, leading to notable improvements in the mechanical properties of the reinforced mortar. In contrast, the 2D auxetic designs, while achieving higher peak stress, exhibit increased brittleness. The ability to tailor auxetic reinforcement configurations based on specific structural requirements highlights their versatility in applications where both strength and ductility are crucial.
- The integration of mixed reinforcement designs – combining auxetic structures in compressive zones and non-auxetic elements in tensile regions – substantially improves the load-bearing performance of mortar beams. This strategy enables tailored deformation behavior, mitigates crack propagation, and ultimately extends the service life of structural components.
- The mechanical performance exhibits notable sensitivity to design parameters such as the re-entrant angle, highlighting the critical role of geometrical optimization in enhancing structural resilience. Compression tests reveal that the strength of both 2D and 3D plate models can be significantly improved by tuning the re-entrant angle, whereas 3D stick models show limited sensitivity due to reduced steel–mortar interaction. Bending tests further demonstrate that increasing the re-entrant angle in mixed designs enhances load-bearing capacity, albeit at the expense of reduced ductility.
- The findings of this study establish a solid foundation for the innovative application of auxetic structures in concrete reinforcement. Auxetic materials show significant potential as alternatives or supplements to conventional reinforcement strategies, particularly in the field of structural engineering. This work contributes to the growing body of knowledge on auxetic design and lays the groundwork for future research aimed at developing more resilient and durable infrastructure.

Future investigations could focus on the integration of auxetic reinforcements across diverse structural systems, evaluation of their behavior under varying loading conditions, and systematic optimization of geometric configurations to maximize mechanical efficiency and structural integrity.

CRediT authorship contribution statement

Yafan Yang: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Xingyu Zhao:** Writing – review & editing, Investigation, Formal analysis. **Cheuk Lun Chow:** Writing – review & editing, Resources, Project administration, Funding acquisition. **Denvin Lau:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The work described in this paper was supported by a grant from the Research Grants Council of the Hong Kong Special Administrative Region, China (Project No. CityU R1018-22). Y.Y. acknowledges the support from the National Natural Science Foundation of China under Grant No. 42203041, the Natural Science Foundation of Jiangsu Province under Grant No. BK20221132, and the Hong Kong Scholars Program (XJ2023042).

Appendix A. Supplementary data

Supplementary figures and videos are given in the supplementary document.

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.tws.2025.113907>.

Data availability

Data will be made available on request.

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